Noise Schemas Aid Hearing in Noise

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Introduction

- Human listening abilities are remarkably robust to noise, but the computational basis of this robustness is poorly understood
- **We explored whether internal models of real-world noise—"noise schemas"— aid the perception of foreground sounds in noise**

Potential Explanations

- **Fixed noise suppression**: fixed filters that attenuate stationary, noise-like sounds
- **Adaptive noise suppression**: noise is suppressed via local adaptation to reduce responses to features that are relatively constant in the auditory input
- **Internal noise modeling**: estimation of ongoing background noise statistics, allowing foreground sounds to be detected as outliers from the associated distribution; properties of familiar noises could be stored over time akin to "schemas" that aid segregation of familiar words or melodies

Stimuli

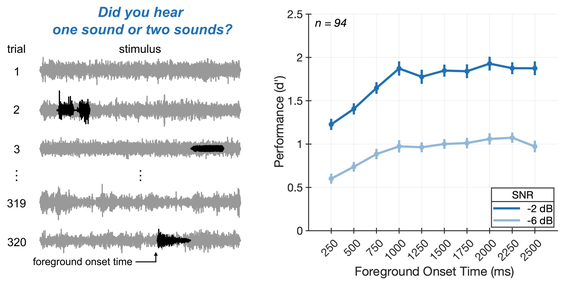
- Diverse set of 160 natural foreground sounds and background noises
- Foreground sounds were 0.5 s excerpts of recorded natural sounds
- Background noises were 3.25 s excerpts of sound textures synthesized from statistics of real-world textures

Conclusions

- Detection, recognition, and localization in noise improved with exposure to the noise
- A model designed to detect outliers from a distribution of background noise accounted for this pattern of human performance
- Performance was enhanced for recurring noises and robust to interruptions in the noise, suggesting representations of noise properties are built up and maintained over time
- **Results suggest noise robustness is supported by internal models—"noise schemas"—that capture the structure of noise and facilitate estimation of other concurrent sounds**

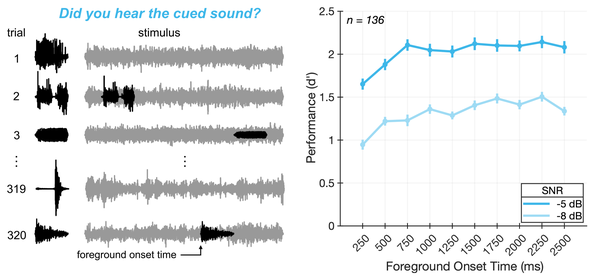
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Experiment 1: Foreground Detection



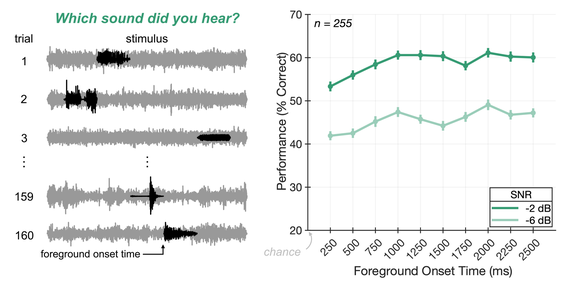
- Detection performance improved with exposure to the background, plateauing after about one second of exposure to the background
- Benefit of background exposure was evident at both SNRs tested

Experiment 2: Cued Detection



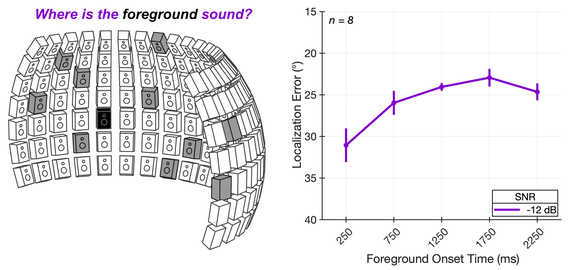
- Cued detection performance also improved with exposure to the background
- Benefit of background exposure persists even when participants know what to listen for

Experiment 3: Foreground Recognition



- Foreground recognition improved with exposure to the background in much the same way as did detection
- Benefit of background exposure extends to sound recognition

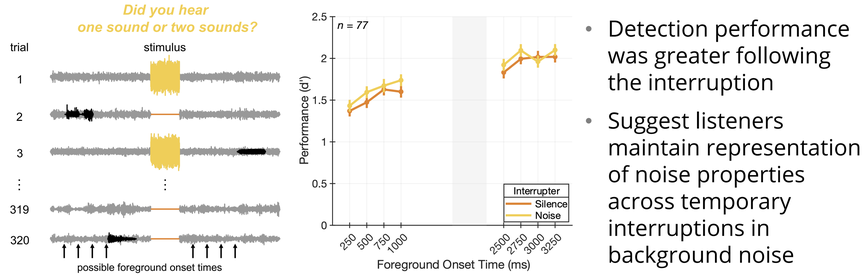
Experiment 4: Foreground Localization



- Foreground localization improved with exposure to the background in a manner similar to that observed for detection and recognition tasks
- Benefit of background exposure extends to sound localization

Experiment 5: Background Interruption

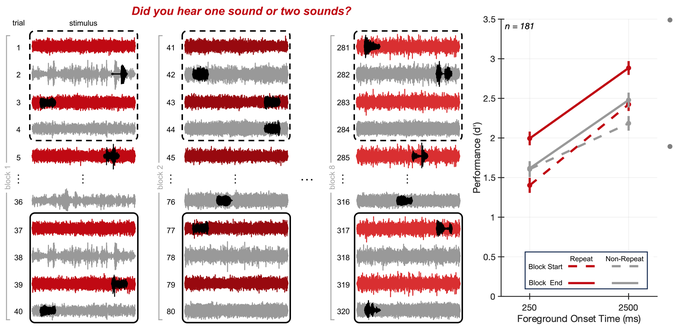
- Interrupted middle 0.5 s of background noise with either ***silence*** or ***white noise***
- This change to the background might cause adaptation to "reset", leading to a decrement in detection performance following the interruption
- The noise schema hypothesis could allow for a benefit of background exposure, since estimated noise properties can be stored across the interruption



- Detection performance was greater following the interruption
- Suggest listeners maintain representation of noise properties across temporary interruptions in background noise

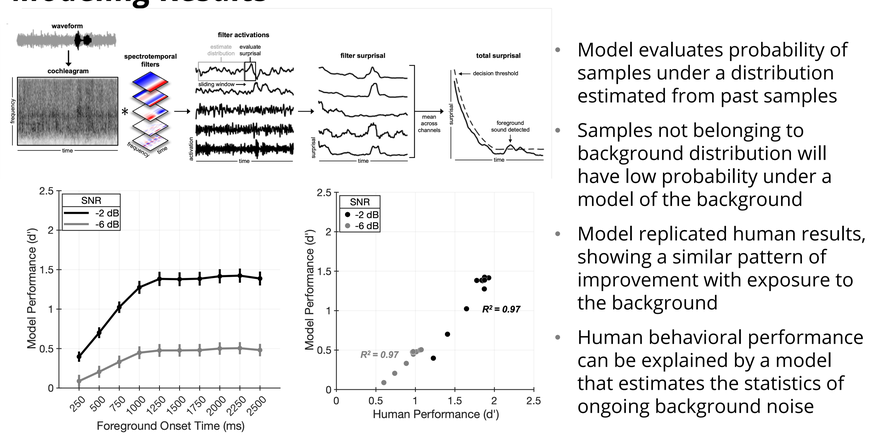
Experiment 6: Repeated Backgrounds

- Are internal models of noise maintained over longer periods of time and across intervening stimuli, akin to "schemas" learned for patterns in speech and music?
- If listeners learn and use noise schemas to aid hearing in noise, detection performance should be enhanced for frequently recurring background noises



- Detection performance was enhanced for recurring background noises
- Suggests that noise schemas (representations of noise statistics that aid detection) are built up and maintained across intervening stimuli

Modeling Results



- Model evaluates probability of samples under a distribution estimated from past samples
- Samples not belonging to background distribution will have low probability under a model of the background
- Model replicated human results, showing a similar pattern of improvement with exposure to the background
- Human behavioral performance can be explained by a model that estimates the statistics of ongoing background noise